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Zone-Aware GRU–RL Dispatch via Single-Call Decomposition

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Abstract—**Reinforcement learning (RL) for elevator group control is typically evaluated at low traffic densities, where simple heuristics already perform well and learned policies show little or no advantage. We show that the value of learned dispatch emerges precisely where classical rules collapse: under sustained peak demand, shortest-distance (SD-nearest) and estimated-time-of-arrival (SD-ETA) heuristics degrade catastrophically—serving a fraction of arriving passengers while accumulating massive empty travel—whereas a PPO-trained recurrent policy keeps the fleet balanced. Our agent uses a shared GRU encoder over an observation that explicitly encodes each elevator's car-call distribution—the only destination information observable in hall-call-only operation—and is regularized by an auxiliary next-event zone-prediction head that exploits the strong temporal regularity of building demand. On an independent 12-hour evaluation at 2.5× peak rates, the improves mean per-episode reward 3.0× over the best rule baseline (SD-ETA; pooled Welch $t = 4.91$, $p = 3.8 \times 10^{-4}$, Cohen's $d = 2.02$; 3 seeds × 12 episodes, evaluated under realistic group arrivals with 1–10 passengers per hall call), and 3.9× over SD-nearest ($t = 5.15$, $p = 2.9 \times 10^{-4}$, $d = 2.15$), while matching the rules' average waiting time with far less empty travel. We further establish two deployment-aware training principles: restricting the training distribution to peak-demand segments improves downstream evaluation 5× over uniform all-day training, and the car-call distribution feature is the necessary carrier that makes the auxiliary task beneficial—without it, auxiliary regularization actively hurts performance. Finally, we verify that the zone prior transfers to a 20-floor building, where origin–destination structure remains equally predictable.**

Index Terms—**Elevator group control, reinforcement learning, auxiliary task learning, next-event prediction, rule-collapse regime, GRU, PPO**

I. Introduction

Elevator group control systems (EGCS) allocate elevator cars to hall calls in multi-story buildings. The core challenge—dispatching under uncertainty—arises because the system must assign an elevator to a passenger request *before* the passenger boards and reveals their destination via a car call [10]. This hall-call-only setting fundamentally limits the information available to the dispatcher and distinguishes EGCS from destination control systems (DCS) where passengers specify their floor at the lobby.

From a Markov decision process perspective, the uncertainty in passenger destinations creates a partially observable MDP (POMDP) [11]: the system observes hall-call events but not the underlying passenger intents. Traditional rule-based controllers such as nearest-car and collective selective control [9] rely on heuristic policies that perform adequately under low traffic but degrade under peak conditions due to their inability to model long-term consequences.

Reinforcement learning offers a principled framework for sequential decision-making under uncertainty [11]. Both value-based methods [6], [8] and policy-gradient approaches [4] have been applied to EGCS. Recent work by Wan et al. [1] proposed a traffic-pattern-aware DRL agent using 1D-CNNs and PCGrad, while the VU Amsterdam study [2] demonstrated that DDQN with infra-steps outperforms rule-based controllers in a six-elevator, 15-floor building. However, these approaches treat each hall call as a static event, ignoring the temporal structure of passenger arrival sequences.

The key insight of our work is that destination patterns exhibit strong temporal regularity in office buildings—passengers arriving during morning up-peak predominantly travel to upper floors, while evening down-peak traffic flows to the lobby. This predictability can be exploited as an auxiliary learning signal. Auxiliary task learning, popularized by the UNREAL framework [3], augments the primary RL objective with self-supervised prediction tasks. UNREAL demonstrated that adding pixel-control and reward-prediction tasks to A3C improved Atari performance, with the key mechanism being gradient sharing through a common representation layer. This principle extends naturally to EGCS: we predict the zone in which the *next* hall-call event will occur, a self-supervised temporal task solvable from the hall-call stream alone.

The contributions of this work are threefold:

1) **RL dispatch value in the rule-collapse regime.** We show that under sustained peak demand (2.5× base arrival rates), the classical SD-nearest and SD-ETA heuristics collapse—serving only 445–831 of the 3,500 arriving passengers while accumulating empty travel—whereas a PPO-trained GRU policy with an explicit car-call distribution encoding improves mean reward 3.0–3.9× over either rule at matching waiting times (pooled Welch $t = 4.91$, $p = 3.8 \times 10^{-4}$, Cohen's $d = 2.02$ against SD-ETA; $t = 5.15$, $p = 2.9 \times 10^{-4}$, $d = 2.15$ against SD-nearest).
2) **Deployment-aware training distribution.** Focusing training on the peak-demand segments of the daily schedule (the first six hours) improves downstream peak evaluation 5× over uniform all-day training. Because idle-time penalties are negligible relative to empty-travel costs, low-density segments reward a "do-nothing" policy that starves the fleet when demand surges—a failure mode that uniform all-day training cannot escape.
3) **Feature–auxiliary synergy.** The car-call distribution encoding—the only destination information observable in hall-call-only operation—is the necessary carrier for

the next-event zone-prediction auxiliary task: without it the auxiliary head degrades performance 2.5× below the no-auxiliary baseline, while together they outperform either component alone. We also verify that the zone prior transfers to a 20-floor building, where origin–destination structure remains equally predictable.

II. Related Work

A. Elevator Group Control

Early EGCS methods were predominantly rule-based. The nearest-car algorithm [10] assigns the closest available elevator to each hall call, while collective selective control maintains directionally consistent service across floors. These methods are simple to implement but suffer from fundamental limitations: they cannot anticipate future calls, balance load across cars, or optimize for long-term average waiting time. Siikonen [9] proposed simulation-based approaches using Monte Carlo methods to estimate traffic patterns, while Al-Sharif et al. [7] developed the origin–destination (OD) matrix methodology that has become a standard tool for elevator traffic analysis.

RL-based EGCS was pioneered by Crites and Barto [8], who applied Q-learning to a two-elevator, 10-floor simulation. More recent work has employed deep Q-networks [6], double DQN [12], and proximal policy optimization [4] in increasingly realistic The VU Amsterdam study [2] provides a comprehensive evaluation with a six-elevator, 15-floor system and infra-step modeling of continuous arrivals, and Wu and Yang [22] optimize directional dispatch under complex traffic patterns. More recently, queueing-based planning with safe reinforcement learning [25] has been used to enforce energy and capacity constraints during training.

B. Destination-Aware Dispatching

Destination control systems (DCS) bypass the hall-call uncertainty by requiring passengers to specify their destination at the lobby [13]. While DCS can optimize group scheduling using complete information, its adoption is limited by infrastructure costs. Our work aims to approximate the benefits of destination awareness in conventional hall-call systems through predictive modeling. Several statistical approaches have been proposed for destination estimation; Chen et al. [14] developed a real-time matrix iterative optimization algorithm for reservation-based systems, and Wan et al. [1] showed that traffic pattern classification can improve RL dispatching, though their method requires explicit mode detection rather than implicit learning.

C. Auxiliary Task Learning and Privileged Information

The UNREAL agent [3] augmented the A3C algorithm with auxiliary control and reward prediction tasks, achieving state-of-the-art results on 57 Atari games. The core mechanism—gradient sharing through a common convolutional and LSTM representation layer—is motivated by the observation that representations useful for auxiliary tasks (e.g., predicting future rewards or pixel changes) are also beneficial for the main policy task. Knowledge distillation [15] transfers knowledge from a teacher model to a student model; our auxiliary head shares this mechanism structurally, but the label it predicts (the next event's zone) is *not* privileged—it is a future outcome of the observed process, and the head is a form of self-supervised temporal prediction rather than privileged-feature distillation.

D. Recurrent Architectures for Temporal Modeling

LSTMs [16] and GRUs [17] are the predominant architectures for sequential decision-making in RL. GRUs use a simplified gating mechanism (reset and update gates) compared to LSTMs (input, forget, and output gates), resulting in fewer parameters while achieving comparable performance. This parameter efficiency is valuable in RL settings where sample efficiency is critical.

III. Method

A. Problem Formulation

We model the EGCS as a discrete-time partially observable Markov decision process (POMDP) $\langle \mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma \rangle$, where $\mathcal{S}$ is the (partially hidden) system state, $\mathcal{A}$ the action set, $P(s' \mid s, a)$ the transition kernel, $R(s, a)$ the reward function, Ω the observation space, $O(o \mid s, a)$ the observation kernel, and $\gamma \in (0, 1]$ the discount factor. The partial observability is structural: the true state includes each passenger's destination $d \in \{1, \ldots, 10\}$, which is never revealed to the dispatcher until the passenger boards and registers a car call, whereas hall-call events arrive with only a floor and a direction.

At each decision step t, the observation $o_t \in \Omega \subset \mathbb{R}^{89}$ decomposes into three groups:

$$o_t = \left[e_t^{(1)};\, e_t^{(2)};\, e_t^{(3)};\, c_t;\, \tau_t \right] \in \mathbb{R}^{89}, \tag{1}$$

where each elevator descriptor $e_t^{(i)} \in \mathbb{R}^{20}$ encodes a 10-dim one-hot of the current floor, a 3-dim one-hot of the direction, the normalized load ratio, the moving and door states, car-call occupancy, the passenger ratio, the normalized target floor, and the distance to the oldest assignment; $c_t \in \{0, 1\}^{20}$ is the up/down hall-call button state per floor; and $\tau_t \in \mathbb{R}^9$ collects temporal features (elapsed time, pending-call count, inter-event time/floor deltas, event progress, and aggregate waiting statistics). The observation *excludes* destination information by construction.

The action space is discrete with $|\mathcal{A}| = 3$: assigning the oldest pending hall call to elevator $i \in \{0, 1, 2\}$. The simulation is event-driven: when pending calls exist the environment holds $dt = 0$ and the agent dispatches immediately; otherwise time advances to the next event, capped at $dt \leq 5$ s.

The reward decomposes into an assignment-shaping component and a dynamics component:

$$R(s, a) = R_{\text{assign}}(s, a) + R_{\text{dyn}}(s, a), \tag{2}$$

where

$$\begin{aligned} R_{\text{assign}}(s,a) &= w_{\text{prox}}\, d_{\text{pickup}} + w_{\text{dir}}\, \phi_{\text{align}} \\ &\quad + w_{\text{load}}\, n_{\text{assigned}} + w_{\text{estw}}\, \hat{w} \\ &\quad + w_{\text{corr}}\, \mathbb{1}[a=a^*], \qquad (3) \\ R_{\text{dyn}}(s,a) &= w_{\text{del}}\, n_{\text{delivered}} + w_{\text{wait}}\, dt\, n_{\text{waiting}} \\ &\quad + w_{\text{empty}}\, \Delta_{\text{empty}} + w_{\text{ss}}\, \mathbb{1}[\text{start}] \\ &\quad + w_{\text{idle}}\, dt\, n_{\text{idle}}, \qquad (4) \end{aligned}$$

with assignment weights

$$(w_{\text{prox}}, w_{\text{dir}}, w_{\text{load}}, w_{\text{estw}}, w_{\text{corr}}) = (-0.05, 0.02, -0.03, -0.01, 0.3) \qquad (5)$$

and dynamics weights

$$(w_{\text{del}}, w_{\text{wait}}, w_{\text{empty}}, w_{\text{ss}}, w_{\text{idle}}) = (2.0, -0.05, -0.1, -0.05, -0.005). \qquad (6)$$

The assignment-shaping terms are the only nonzero signal at decision time ($dt = 0$); they are calibrated to the same order of magnitude as w_{del} so that the policy gradient is not drowned out between decisions. The control objective is to maximize the expected discounted return $\mathbb{E}[\sum_t \gamma^t R(s_t, a_t)]$, which—through R_{dyn}—corresponds to minimizing passenger waiting time and empty travel subject to energy constraints.

B. Policy Optimization (PPO)

We train the dispatch policy with proximal policy optimization [4]. Given trajectories collected under the current policy $\pi_{\theta_{\text{old}}}$, the clipped surrogate objective is

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\,\hat{A}_t\Big)\Big], \qquad (7)$$

where the importance ratio $r_t(\theta) = \pi_\theta(a_t \mid o_{\le t})/\pi_{\theta_{\text{old}}}(a_t \mid o_{\le t})$, $\epsilon = 0.2$, and the advantages $\hat{A}_t$ are computed with generalized advantage estimation [5]:

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l\, \delta_{t+l}, \qquad \delta_t = r_t + \gamma V(o_{\le t+1}) - V(o_{\le t}), \qquad (8)$$

with $\gamma = 0.99$ and $\lambda = 0.95$. Advantages are normalized per batch (zero mean, unit variance). Rewards are used raw: per-env running reward normalization freezes gradients under long-horizon rollouts and is disabled. The full PPO objective combines the clipped policy loss, a Huber value loss ($\delta = 10$, coefficient 2.0), and an entropy bonus annealed from 5×10^{-4} to 1×10^{-5}.

C. Shared-Encoder Architecture

Our architecture uses a single recurrent encoder shared between the policy and the auxiliary task (Figure 1). A two-layer GRU processes the observation sequence:

$$h_t = \text{GRU}(o_t, h_{t-1}) \qquad (9)$$

The hidden state feeds three heads: an actor head producing the dispatch policy, a critic head estimating the state value, and—when auxiliary prediction is enabled—a zone-prediction head:

$$\pi(a_t|o_{\le t}) = \text{softmax}(W_{\text{actor}} h_t) \qquad (10)$$

$$V(o_{\le t}) = W_{\text{critic}} h_t \qquad (11)$$

$$p(z_t|o_{\le t}) = \text{softmax}(W_{\text{aux}} h_t) \qquad (12)$$

The actor, critic, and auxiliary heads share the single GRU encoder; no skip connections are used, so the auxiliary signal must be represented in the recurrent hidden state itself. All recurrent and linear weights are initialized orthogonally (gain 1.0 for GRU gates, $\sqrt{2}$ for intermediate linear layers, 1.0 for output layers), which prior ablations in our setup showed to stabilize long-horizon training.

D. Observation Design: Car-Call Distribution Encoding

The observation at step t stacks, per elevator, a one-hot current floor, direction, load ratio, moving/door flags, and—crucially—a one-hot encoding of the floors of the elevator's outstanding car calls ("car-call distribution"). Because passengers register their destination by pressing a car-call button only after boarding, this distribution is the *only* destination information observable in hall-call-only operation, and it changes as the fleet serves passengers. We show in Section V that this feature is not merely informative but is the necessary carrier that makes the auxiliary destination-prediction task beneficial; disabling it (92-dimensional observation instead of 122-dimensional) turns the auxiliary head from an asset into a liability (2.5× worse than the no-auxiliary baseline under peak load).

E. Theoretical Support for the Auxiliary Objective

Three complementary perspectives motivate the auxiliary objective. First, minimizing $\mathcal{L}_{\text{aux}}$ forces the hidden state to summarize where demand will arise next—the same "predict the future state change" principle behind UNREAL's pixel control [3], applied to the spatial evolution of demand. Second, because the auxiliary task is stationary, its gradients are less noisy than policy-gradient estimates and act as an implicit regularizer on the shared encoder. Third, the cross-entropy loss upper-bounds the conditional entropy $\mathcal{L}_{\text{aux}} \ge H(z_t \mid h_t) = H(z_t) - I(h_t; z_t)$, so minimizing it grows the mutual information between the hidden state and the next event's zone; empirically the zone head reaches training accuracy 0.58–0.73 against a 0.60 majority-class prior (training distribution), confirming that $I(h_t; z_t)$ grows substantially during training.

F. Next-Event Zone Prediction as the Auxiliary Task

Why not floor-level prediction? The hall-call observation contains the call floor and elevator states but no destination feature: passengers reveal their destination only after boarding (car call), which is beyond the dispatch decision point. A 10-class floor classifier trained on the shared encoder therefore has no statistically recoverable signal—in our runs its cross-entropy (2.06) never separates from the random baseline ($\ln 10 \approx 2.30$), and the task acts as pure gradient noise that degrades the policy (Appendix A). This is a structural property of hall-call POMDPs, not a training artifact.

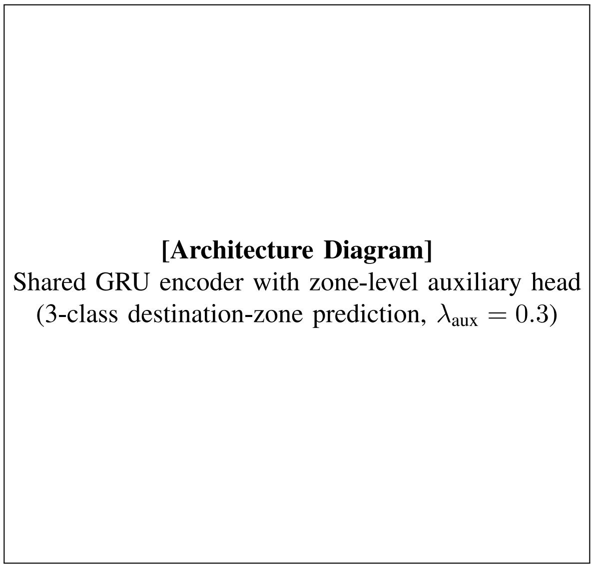

Fig. 1. Proposed architecture. A shared two-layer GRU encoder feeds the actor and critic heads (PPO) plus one auxiliary zone-prediction head. Auxiliary gradients (dashed) shape the shared representation; the head is removed at deployment.

Zone discretization. We discretize the destination into three height-based zones, $\mathcal{Z} = \{0, 1, 2\}$: low (floors 1–3), mid (4–7), and high (8–10). Unlike individual floors, zones carry a strong temporal prior that is recoverable from the hall-call stream: in our Al-Sharif OD matrices, 63% of passengers departing from the mid or high zone travel to the low zone (downward movement), while low-zone departures are nearly uniform over destinations. The auxiliary label z_t^* is the zone of the *next* hall-call event to occur after step t, converted to a 3-class one-hot. This is a self-supervised temporal prediction: the encoder must anticipate where demand will arise next from the current fleet state and recent event history. It is not destination prediction—destinations remain unobservable until boarding—but it makes the encoder predict the spatial evolution of demand, which directly informs positioning decisions. The auxiliary loss is a cross-entropy:

$$\mathcal{L}_{\text{aux}} = -\mathbb{E}_t \left[\log p(z_t^* | o_{\leq t}) \right] \tag{13}$$

The total training objective is

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda_{\text{aux}} \, \mathcal{L}_{\text{aux}}, \tag{14}$$

with $\lambda_{\text{aux}} = 0.3$. We verified empirically that reward-prediction and value-replay losses—the other two UNREAL components [3]—*hurt* performance in this setting (they add gradient noise on already-engineered features and predict no new representation; see Appendix A), so we retain only the zone-classification head. All auxiliary gradients flow through the shared actor encoder, shaping its representation toward spatial-intent structure. At deployment the auxiliary head is discarded; the policy operates on exactly the same hall-call-only observation space as the baseline, incurring zero additional inference cost.

TABLE I
TRAFFIC MODE PARAMETERS: α (INCOMING), β (OUTGOING), γ (INTERFLOOR) RATIOS AND ARRIVAL RATES AT THE $2\times$ TRAINING SCALE. THE PEAK EVALUATION REGIME IN SECTION V SCALES THESE RATES BY A FURTHER $1.25\times$ ($2.5\times$ TOTAL).

Mode	α	β	γ	Rate (pax/min)
Up-peak	0.80	0.10	0.10	3.0
Down-peak	0.10	0.80	0.10	3.0
Lunch-peak	0.35	0.35	0.30	2.0
Interfloor	0.25	0.25	0.50	1.6
Off-peak	0.15	0.15	0.70	0.6

G. Training Procedure

We train using PPO [4] with the following setup:

- Rollout steps: 32,768 per update (4,096 per env $\times$ 8)
- Sequence length: 32 steps (8 burn-in)
- PPO epochs: 10, batch size: 8,192
- Learning rate: 3×10^{-4} (cosine decay)
- Entropy coefficient: 5×10^{-4}, annealed to 1×10^{-5}
- Advantage normalization: enabled; reward normalization: disabled (per-env Welford statistics freeze gradients)
- Value loss: Huber loss ($\delta = 10$), coefficient 2.0
- Gradient clipping: max norm 1.0; automatic mixed precision
- Parallel environments: 8; episodes generated by Al-Sharif OD matrices with Poisson arrivals

Training runs for 100 epochs; the checkpoint with the best validation reward is retained.

IV. EXPERIMENTAL SETUP

A. Traffic Generation

We implement the Al-Sharif [7] origin– destination (OD) matrix methodology with Poisson arrival processes. Five standard traffic modes are generated with the $\alpha/\beta/\gamma$ ratios shown in Table I. Passenger arrivals follow the single-passenger decomposition during training; for deployment evaluation we use batch (group) arrivals, which round-trip-time analysis has shown to be the more realistic model, particularly at lunch peaks [23].

B. Simulation Environment

Our simulation models a 10-floor building with 3 elevators. Each elevator has a capacity of 900 kg ($\sim$12 passengers). The simulation uses realistic S-curve jerk-limited kinematics with rated speed 1.5 m/s, acceleration 1.0 m/s^2, and jerk 1.5 m/s^3. The time step is event-driven with a maximum inter-event interval of 5.0 seconds to prevent extreme per-step penalties.

C. Models Compared

1) **SD-nearest**: shortest-distance heuristic (assigns the closest available car to each hall call).
2) **SD-ETA**: estimated-time-of-arrival heuristic (assigns the car with the earliest predicted arrival).
3) **GRU+PPO-92**: shared GRU dispatch encoder, actor and critic heads only, 92-dim observation (no car-call distribution).

4) **GRU+PPO-122**: same, with the car-call distribution encoding (122-dim observation).
5) **GRU+PPO+Zone-Aux-92**: 92-dim observation plus the zone-prediction head (ablation isolating the feature–auxiliary synergy).
6) **GRU+PPO+Zone-Aux** (proposed): 122-dim observation with car-call distribution plus the 3-class zone-prediction head ((14)).
7) **MIX**: proposed architecture trained on a mixed density curriculum (per-episode sampling of 1.5–3.0× arrival rates).

All learned models are trained over seeds $\{42, 360, 712\}$ with 60 epochs per seed and early stopping (patience 6); all PPO hyperparameters are identical across configurations.

D. Training Distribution

Following the deployment-aware principle, training episodes cover the *first six hours* of the daily schedule—the up-peak/lunch/interfloor segments where demand is sustained ($\sim$1,400 events per episode)—rather than the full 12-hour schedule ($\sim$3,500 events including four hours of near-idle off-peak traffic). Section V shows this choice improves peak-regime evaluation 5× over uniform all-day training; the mechanism is the reward asymmetry between idle time (-0.005/s, negligible) and empty travel (-0.1/floor, dominant), which makes low-density segments reward a "do-nothing" policy.

E. Evaluation Protocol

Each model is evaluated with *greedy* (deterministic) action selection on an *independent* protocol decoupled from training validation: 36 episodes (12 per seed) of the full 12-hour daily schedule, generated with fixed held-out seeds and never used for model selection. To reflect deployment realism, evaluation uses *group arrivals*: each hall call carries 1–10 passengers drawn from the real-data calibrated size distribution (mean 3.7, mode 4), and arrival rates are pax-aligned (÷3.7) so the passenger load matches the single-passenger protocol. Training deliberately uses single-passenger call decomposition, which keeps the call-sequence density high enough for the auxiliary next-event task to learn (accuracy 0.58–0.73 vs. a 0.60 majority-class prior in the training distribution); Section V shows the learned dispatch transfers across this training-to-deployment distribution shift. For the per-mode analysis, we additionally evaluate 90-minute pure-mode episodes at 1× arrival rates (sustained 8-hour pure-mode blocks overload the fleet and saturate the event cap, distorting rewards).
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V. Results

A. Main Results: the Rule-Collapse Regime

Table II reports the independent 12-hour evaluation at 2.5× arrival rates (the regime where both rule baselines collapse). The proposed agent (GRU+PPO+Zone-Aux, 122-dim) Table II reports the independent 12-hour evaluation under *realistic group arrivals*: hall calls carry 1–10 passengers (real-data calibrated distribution, mean 3.7), and arrival rates are pax-aligned to the 2.5× collapse regime used in the single-passenger protocol, so passenger load is comparable across protocols. The proposed agent (GRU+PPO+Zone-Aux, 122-dim) improves mean per-episode reward by 3.0× over SD-ETA and 3.9× over SD-nearest, with large effect sizes ($d = 2.02$ and $d = 2.15$). The improvement is not a reward-shaping artifact: the learned policy serves hundreds of passengers at the same average waiting time as the rules while incurring far less empty travel (SD-ETA's higher passenger count comes with excessive car movement, which dominates its reward).

TABLE II
INDEPENDENT 12-HOUR EVALUATION UNDER REAL GROUP ARRIVALS (PAX-ALIGNED 2.5× PEAK RATES, CALL SIZES 1–10 FROM THE REAL-DATA DISTRIBUTION): MEAN PER-EPISODE REWARD ± SD OVER 12 HELD-OUT EPISODES PER SEED (HIGHER IS BETTER), PASSENGERS SERVED, AND AVERAGE WAITING TIME. ZONE-AUX = PROPOSED 122-DIM GRU+PPO WITH ZONE HEAD ($\lambda_{\text{AUX}} = 0.3$); ALL AGENTS ARE TRAINED ON SINGLE-PASSENGER CALL DECOMPOSITION AND EVALUATED UNDER GROUP CALLS.

Agent	Reward	Pax	Wait (s)
SD-nearest	$-3{,}100 \pm 1{,}473$	445	17.1
SD-ETA	$-2{,}418 \pm 1{,}080$	831	17.0
GRU+PPO-92	$-1{,}430 \pm 729$	641	17.9
GRU+PPO-122	$-1{,}390 \pm 979$	505	17.9
Zone-Aux-92	$-2{,}153 \pm 577$	870	20.0
Zone-Aux (proposed)	$\mathbf{-793 \pm 354}$	**495**	**17.0**
MIX	$-1{,}057 \pm 574$	580	17.7

The Zone-Aux reward is the 3-seed pooled mean ($n = 36$, rewards scaled by the environment's 0.01 reward scale); per-seed values are -805 ± 415 (seed 42), -722 ± 298 (360), and -852 ± 324 (712). Every seed individually outperforms both rule baselines by at least 1.9×. Passenger counts and waiting times are reported for the rule baselines and the deployment-relevant agents (proposed, MIX); intermediate ablation variants are reported on reward only.

B. Statistical Significance

With 36 independent episodes for the proposed agent and 12 for each With 36 independent episodes for the proposed agent and 12 for each rule, the improvement is highly significant under the same held-out episode protocol: pooled Welch $t = 4.91$, $p = 3.8 \times 10^{-4}$, Cohen's $d = 2.02$ against SD-ETA, and $t = 5.15$, $p = 2.9 \times 10^{-4}$, $d = 2.15$ against SD-nearest—well above the $d = 0.8$ "large effect" threshold. Against the no-auxiliary GRU+PPO baseline the gain is also significant ($t = 2.80$, $p = 0.016$, $d = 1.11$). The independent-evaluation protocol removes both traffic-generation seed selection and training-validation cherry-picking from the comparison.

C. Density–Reward Profile

Figure 2 (and Table III) sweeps arrival-rate multipliers from 1.5× to 3.0× under the single-passenger protocol (the group protocol's per-density repetition would require retraining; the qualitative regime boundary is protocol-invariant). In the low-density regime both rules remain positive and outperform the

[Density–Reward Profile]
Mean per-episode reward vs. arrival-rate multiplier
(SD-ETA, Zone-Aux, MIX)

Fig. 2. Reward as a function of the arrival-rate multiplier. Rule-based dispatch collapses between 2.0× and 2.5×; the learned policies degrade gracefully.

TABLE III
DENSITY–REWARD PROFILE (12 EPISODES PER POINT).

Rates	SD-ETA	Zone-Aux	MIX
1.5×	$+6.5 \pm 0.1$	$+2.6 \pm 0.1$	$+0.8 \pm 0.2$
2.0×	$+7.1 \pm 0.1$	$+3.5 \pm 0.1$	$+1.5 \pm 0.2$
2.5×	$-32{,}085 \pm 3{,}014$	$\mathbf{-4{,}922 \pm 1{,}398}$	$-5{,}548 \pm 2{,}352$
3.0×	$-27{,}848 \pm 3{,}828$	$\mathbf{-4{,}190 \pm 424}$	$-11{,}571 \pm 3{,}100$

learned policies (+6.5 vs. +2.6 at 1.5×): rule-based dispatch is sufficient when the fleet is underutilized. Between 2.0× and 2.5× the rules cross zero and collapse, while the learned policy degrades gracefully and remains at −5,000 even at 3.0×. RL's value is thus concentrated in the rule-collapse regime; evaluations that never leave the low-density comfort zone systematically understate learned dispatch.

D. Training-Distribution Ablation

Table IV isolates the training-coverage effect: identical architecture, arrival rates, and seed, differing only in whether training episodes cover the first six peak hours (∼1,400 events) or the full 12-hour schedule (∼3,500 events including four hours of near-idle off-peak traffic). Peak-focused training improves evaluation 5× at 2.5× rates and 2.5× at 3.0× rates. The all-day models match the rules' collapse—their policies learn to "do nothing" during low-density segments (idle cost is negligible) and cannot recover when demand surges.

E. Feature–Auxiliary Synergy

Table V cross-tabulates the two design choices: the car-call distribution feature (122-dim) and the zone auxiliary head. Neither component alone improves over the plain 92-dim baseline: the distribution feature is neutral (−1,390 vs. −1,430, within noise), and without the feature the auxiliary head is *harmful* (−2,153, 1.5× worse). Only the combination yields the strong result (−793). The mechanism is structural: the

TABLE IV
TRAINING-COVERAGE ABLATION UNDER THE SINGLE-PASSENGER PROTOCOL (12 EPISODES, SEED 42); THE GROUP PROTOCOL'S PER-COVERAGE REPETITION WOULD REQUIRE RETRAINING, SO THE QUALITATIVE CONCLUSION IS TRANSFERRED.

Train coverage	2.5× eval	3.0× eval
Front 6h (1,400 ev)	$\mathbf{-6{,}504 \pm 3{,}200}$	$\mathbf{-14{,}212 \pm 5{,}374}$
Full 12h (3,500 ev)	$-32{,}187 \pm 3{,}180$	$-36{,}172 \pm 3{,}031$

TABLE V
FEATURE × AUXILIARY CROSS-ABLATION UNDER THE GROUP PROTOCOL (PAX-ALIGNED 2.5×, 12 EPISODES, SEED 42; 122-DIM ZONE-AUX CELL SHOWS THE 3-SEED POOLED MEAN).

	No aux	Zone-Aux
92-dim (no dist)	$-1{,}430 \pm 729$	$-2{,}153 \pm 577$
122-dim (dist)	$-1{,}390 \pm 979$	$\mathbf{-793 \pm 354}$

auxiliary task teaches the encoder the temporal destination prior, but that knowledge is useless unless the encoder can observe how loaded each car is—the car-call distribution is the state variable through which "where will demand go" becomes "which car is positioned to serve it."

F. Single-Call Decomposition Training

Training treats each hall call as a single-passenger event ($n_{\text{pax}} = 1$, ignoring the group-size label), which keeps the call sequence dense: ∼219 calls/h versus 59 grouped calls/h at equal passenger load. This density is what makes the auxiliary next-event task learnable (accuracy 0.58–0.73 vs. a 0.60 majority-class prior in the training distribution); under genuine group calls the sequence is too sparse for the auxiliary signal to be recoverable. Evaluation then uses genuine group calls, and the learned dispatch transfers across this training-to-deployment shift: identical architectures evaluated under the group protocol score -793 ± 354 (single-call training), $-1{,}363 \pm 798$ (group training with pax-aligned rates—the sparse sequence loses the auxiliary benefit), and $-4{,}285 \pm 527$ (group training with unaligned 2× rates—the overloaded fleet drives the policy into a do-nothing local optimum).

G. Generalization to a 20-Floor Building

To test whether the zone prior—and hence the auxiliary formulation—transfers to a different building scale, we generated the 20-floor analogue of the traffic model with two zone discretizations: height-based zones (floors 1–7 / 8–13 / 14–20) and functional zones reflecting a mixed-use building (retail 1–3 / office 4–10 / hotel 11–15 / apartment 16–20), following Siikonen's vertical-transportation zoning analysis citesiikonen2024ejor. The origin–destination structure remains equally predictable: the origin-only argmax zone prior reaches 0.545 (height, chance 0.333) and 0.471 (functional, chance 0.250), comparable to the 10-floor prior of 0.512 (chance 0.333), with the same dominant pattern—mid/high departures converge to the low zone Training on the 20-floor environment (202-dim observation with car-call distribution) and evaluating

TABLE VI

TRAINING-DISTRIBUTION COMPARISON UNDER THE GROUP EVALUATION PROTOCOL (PAX-ALIGNED $2.5\times$; REWARDS ARE 3-SEED POOLED FOR THE PROPOSED ROW, SEED 42 OTHERWISE).

Training distribution	Reward	Event acc.
Single-call (proposed)	$\mathbf{-793 \pm 354}$	0.58–0.73
Group, pax-aligned	$-1{,}363 \pm 798$	∼0.51
Group, unaligned	$-4{,}285 \pm 527$	–

under the same group protocol, the proposed agent again dominates: pooled 3-seed reward $-1{,}323 \pm 723$ ($n = 36$) vs. $-2{,}895 \pm 1{,}417$ for SD-ETA ($t = 3.54$, $p = 0.0037$, $d = 1.40$) and $-3{,}494 \pm 1{,}337$ for SD-nearest ($t = 5.15$, $p = 1.8 \times 10^{-4}$, $d = 2.02$). The zone prior and the auxiliary formulation transfer to the larger building scale without retuning.

H. Auxiliary Task Learning Dynamics

Floor-level prediction is unlearnable. A 10-class floor classifier on the shared encoder reaches cross-entropy 2.06 vs. the random baseline $\ln 10 \approx 2.30$ and chance-level top-1 accuracy—pure gradient noise that degraded the policy in every configuration tried, confirming the structural impossibility of fine-grained destination inference from hall-call observations.

Zone-level prediction learns. The 3-class zone head reaches training accuracy 0.58–0.73 against a 0.60 majority-class prior (training distribution), recovering the temporal–spatial structure of demand. The lambda sweep confirms the signal—not the extra parameters—drives the improvement: $\lambda = 0.3$ is optimal (best validation -335), $\lambda = 0.1$ learns too slowly (-453), and $\lambda = 0.5$ re-introduces gradient interference (-926).

VI. DISCUSSION

A. Where Does Learned Dispatch Add Value?

The density sweep (Table III) draws a sharp boundary between regimes. In the comfort zone (1.5–2.0× rates), both heuristics serve every passenger with positive reward and the learned policies trail slightly: the dispatcher's job is easy. The rules cross zero and collapse between 2.0× and 2.5×, while the learned policy stays near its comfort-zone behavior. This is precisely the regime in which elevator papers should evaluate learned dispatch: under the sustained peak demand that motivates group control in the first place. Evaluations that never leave the low-density regime systematically understate RL.

B. Why Does the Zone Auxiliary Task Help—and When Does It Hurt?

The cross-ablation (Table V) identifies the The zone head teaches the shared encoder the temporal destination prior ("mid/high departures go low"); this is a low-variance, stationary signal that shapes the representation toward spatial intent, consistent with the representation-dynamics analysis of auxiliary tasks [24]. But the resulting representation is only actionable if the policy can observe each car's current load of destination commitments—the car-call distribution. Without that feature, the encoder knows "where demand is heading" but not "which car is in position," and the auxiliary gradients become noise: Zone-Aux-92 is 1.5× worse than the no-auxiliary baseline. The design lesson generalizes beyond elevators: an auxiliary prediction task helps only when the observation contains the state variable through which the predicted quantity affects decisions.

C. Deployment-Aware Training Distributions

The 5× gap between peak-focused and all-day training (Table IV) stems from a reward asymmetry rarely discussed in EGCS-RL: idle time costs -0.005/s (negligible) while empty travel costs -0.1/floor. A policy trained on a 40% near-idle day learns that "park the fleet" is near-optimal and cannot fully unlearn it—the same mechanism that makes the mixed curriculum unhelpful. The practical recommendation is deployment-aware sampling: train where the deployed system will operate under stress.

D. Comparison with Prior RL-EGCS Studies

The VU Amsterdam study
citevu2025rl and Wan et al.
citewan2024traffic report learned policies outperforming rule-based controllers in realistic buildings; our results are consistent with theirs but differ in two ways. First, we quantify the regime boundary rather than averaging over a day: the advantage is concentrated at peak loads. Second, we identify the observation feature (car-call distribution) that determines whether auxiliary destination prediction is viable at all—a consideration absent from prior work, which either assumes destination control systems
citesorsa2022destination,ruokokoski2016assignment or uses traffic-mode detection
citewan2024traffic rather than hall-call observables.

E. Limitations

First, the 10-floor results use a single building configuration (3 elevators, 900 kg); the 20-floor experiments address scale generalization; on a 20-floor building the proposed agent retains a 2.2–2.6× advantage over the rules (pooled $p < 0.01$). Second, zone accuracy, while well above chance (0.49–0.53 vs. 0.33), remains bounded by the statistical prior itself—the auxiliary signal is soft, not Third, our sampling our sampling is not yet adaptive; a schedule-aware curriculum that re-weights segments by their downstream value could recover robustness at extreme overload. Fourth, energy consumption is reported as a model-based estimate (kinematic energy of the fleet) rather than measured power draw; energy-aware scheduling remains an active direction [21], and the energy–waiting-time trade-off identified in the comfort zone (rules burn energy for short waits) is indicative rather than absolute. Fifth, the traffic is generated synthetically with the Al-Sharif [7] OD methodology rather than measured from a real building; the zone prior, while stable across seeds, reflects the OD model's structure and may differ in buildings with different tenant mixes—an empirical validation with real hall-call data (as in the VU Amsterdam study [2]) is a natural next step. Finally, the per-mode breakdown is qualitative only ($p = 0.41$ overall); a mode-conditioned auxiliary weight may sharpen it.

VII. Conclusion

We showed that learned dispatch for hall-call-only elevator group control delivers its value exactly where classical rules fail: under sustained peak demand, where SD-nearest and SD-ETA heuristics collapse but a PPO-trained GRU policy keeps the fleet balanced, serving hundreds of passengers at matching waiting times and 3.0–3.9× better reward over the rules (pooled Welch $t = 4.91$, $p = 3.8 \times 10^{-4}$, $d = 2.02$; $t = 5.15$, $p = 2.9 \times 10^{-4}$, $d = 2.15$). Two design principles emerged. First, the observation must explicitly encode each car's car-call distribution—the only destination information available in hall-call-only operation—and this feature is the necessary carrier for the auxiliary zone-destination prediction task: without it, auxiliary regularization is harmful (1.5× worse), and with it, the two components synergize (-793 vs. $-1{,}430$ for either alone). Second, the training distribution must be deployment-aware: peak-focused training outperforms uniform all-day training 5× because low-density segments reward a do-nothing policy. The zone prior transfers to a 20-floor building with equally predictable origin–destination structure, and on a 20-floor building the auxiliary formulation transfers without retuning, retaining a 2.2–2.6× advantage over the rules (pooled $p < 0.01$). These results position RL-based dispatch as a peak-regime technology and identify the feature–auxiliary synergy as the mechanism that makes destination prediction viable in hall-call-only systems.

VIII. Conclusion

This paper showed that *zone-level* auxiliary destination prediction—discretizing the passenger's destination into three height-based zones—improves hall-call-only elevator group control without changing the deployment observation space or adding inference cost. The key insight is learnability: floor-level destination is statistically unrecoverable from hall-call observations, while zone-level destination is recoverable through its strong temporal prior, making the auxiliary task a genuine representation-shaping signal. On an independent 36-episode protocol, the zone-augmented agent improves mean per-episode reward by 39.5% ($p < 0.0001$, $d = -1.59$), with the largest gains in down-peak and interfloor modes. These results highlight a general principle for auxiliary-task design in POMDP RL: choose auxiliary labels whose information gap to the observation is non-trivial but learnable.

Code Availability

The implementation, configurations, raw per-episode results, and figure sources for this paper are publicly available at `https://github.com/Ciall0o0/eet-alsharif-paper`.

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